

SUFFOLK COUNTY COMMUNITY COLLEGE
ANNUAL REPORT OF ACADEMIC ASSESSMENT ACTIVITY

Thank you for completing the academic assessment activity report. This documentation is important in demonstrating our commitment to using student learning assessment to improve teaching and learning.

Academic Year reporting on: Year # in 5-Year Assessment Plan: **Fall 2024 - Spring 2025**

There are three sections to this report:

Section I: Report on assessment activity follow-up from the previous Academic Year:

Section II: Report on current assessment activity for this Academic Year:

Section III: Report on Program Review Recommendations.

ANNUAL REPORT OF ACADEMIC ASSESSMENT ACTIVITY

Academic Program and Degree:	Liberal Arts and Sciences: Science – Earth and Space Science – Astronomy Option (LAAS-AS)	
Lead person(s):	For this current report using the standardized assessment instrument ADT: Glenda Denicolo Further assessments in astronomy and previous years: Matthew Pappas	
Date of Assessment Report:	May 30, 2025	
Does this PO align with a SUNY General Education learning outcome?	NO If YES , please indicate which one: The PO assessed in this report is PLO#1 (“ <i>Demonstrate knowledge of factual material essential to their discipline in science.</i> ”), which aligns with the following SUNY GE learning outcomes: <i>Mathematics:</i> ○ <i>interpret and draw inferences from appropriate mathematical models such as formulas, graphs, tables, or schematics;</i> ○ <i>represent mathematical information symbolically, visually, numerically, or verbally as appropriate; and</i> ○ <i>employ quantitative methods such as arithmetic, algebra, geometry, or statistics to solve problems.</i> <i>Natural Sciences:</i> ○ <i>an understanding of the methods scientists use to explore natural phenomena, including observation, hypothesis development, measurement and data collection, experimentation, evaluation of evidence, and employment of data analysis or mathematical modeling; and</i> ○ <i>application of scientific data, concepts, and models in one of the natural sciences.</i> <i>Critical Thinking and Reasoning:</i>	

	○ <i>clearly articulate an issue or problem;</i> ○ <i>identify, analyze, and evaluate ideas, data, and arguments as they occur in their own or others' work;</i> ○ <i>acknowledge limitations such as perspective and bias; and</i> ○ <i>develop well-reasoned (logical) arguments to form judgments and/or draw conclusions.</i>
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Section I: Report on assessment activity follow-up from Academic Year Fall 2023/Spring 2024.

What Program Learning Outcome(s) did you assess last academic year?	Program Learning Outcome #1: <i>Demonstrate knowledge of factual material essential to their discipline in science.</i>
Describe any of actions that were taken this year in response to the assessment findings. Please give specific examples of any changes you have made as a result of the findings.	The importance of distinguishing between the different exoplanet search programs (CLO A) was evaluated and determined to be a valuable way to have students evaluate how our knowledge of exoplanets depends on the technology available to conduct research. As new search programs come online more details about exoplanets will emerge that weren't previously known. CLO A will thus remain as a learning outcome for the course. The option of embedding assessment questions will be discussed at a future meeting of astronomy faculty and voted upon in time for the next scheduled offering of the course in Spring 2026. Reevaluation of course materials and activities is ongoing and recommended changes will be addressed at a future meeting of astronomy faculty in time for the next scheduled offering of the course in Spring 2026.

Section II:

Report on completed assessment activity for Current Academic Year: Fall 2024/Spring 2025.

What Program Learning Outcome(s) did you assess?	Program Learning Outcome #1: <i>Demonstrate knowledge of factual material essential to their discipline in science.</i>
Description of assessment activity. Please include (1) the methodology, as well as any specific measurement criteria; (2) the course(s) the assessment took place. (3) the desired student performance from (4) the number of	We have applied the research-based assessment tool called Astronomy Diagnostic Test 2.0 (ADT) to a total of six sections of AST101 (3 sections) and AST102 (3 sections) in the past academic year. Both AST101 and AST102 received the same questions, the same assessment instrument. ADT, its resources and references can be found at https://www.physport.org/assessments/. The content knowledge covered in ADT includes mainly: apparent motion of the Sun, scale of the solar system, phases of the moon, linear distance scales, seasons, global warming, nature of light, gravity, stars, cosmology. Information on how to obtain the test is attached at the end. (1) Methodology: ADT was applied on the first day of classes (pre-test) and on the last week of the semester (post-test). Students were given 35 minutes to complete the test, but most completed it faster than that. Only 21 questions relate to the subject of astronomy, making 21 as the maximum score. The remaining

students who participated in the assessment.

questions addressed confidence level, highest math-level taken, ethnicity, expectations.

(2) Courses where ADT was applied:

Semester	Course - CRN	Number of students matching Pre/Post tests	Instructor
Fall 2024	AST101 - 90833	15	Glenda Denicolo
	AST101 - 95844	12	
	AST102 - 95204	12	
Spring 2025	AST101 - 20743	22	
	AST102 - 35523	20	
	AST102 - 33955	10	
Total:		91	

(3) Desired student performance:

Our objective is to compare our results with ADT national statistics presented below for the *Lecture + lab* format:

	Lecture	Lecture + discussion	Lecture + lab	Lecture + lab + discussion
# classes	16	5	40	9
# students pretest	1045	549	1730	723
Mean prescore (21 max)	6.65	6.09	6.70	7.31
Standard deviation	.71	.28	1.09	.90
# students posttest	758	369	1371	582
Mean postscore (21 max)	9.66	8.60	9.77	11.44
Standard deviation	1.61	1.13	1.88	2.01
Mean normalized gain	.2098	.1681	.2146	.3016
Standard deviation on normalized gain	.0870	.0628	.0973	.1221
Standard error on normalized gain	.0217	.0281	.0154	.0407

Source: Brogt et al., 2007. <https://doi.org/10.3847/AER2007003>

Following the literature, we have also used normalized gain (g) and the effect size (d) as metrics for comparison.

$$\text{Normalized gain: } g = \frac{\langle \text{post} \rangle - \langle \text{pre} \rangle}{(\text{max score}) - \langle \text{pre} \rangle}$$

<https://www.physport.org/recommendations/Entry.cfm?ID=93334>

$$\text{The effect size: } d = \frac{\langle \text{post} \rangle - \langle \text{pre} \rangle}{\text{std dev}}$$

<https://www.physport.org/recommendations/Entry.cfm?ID=93385>

(4) Total number of students who participated in the assessment: 91 students (49 from AST101, 42 from AST102).

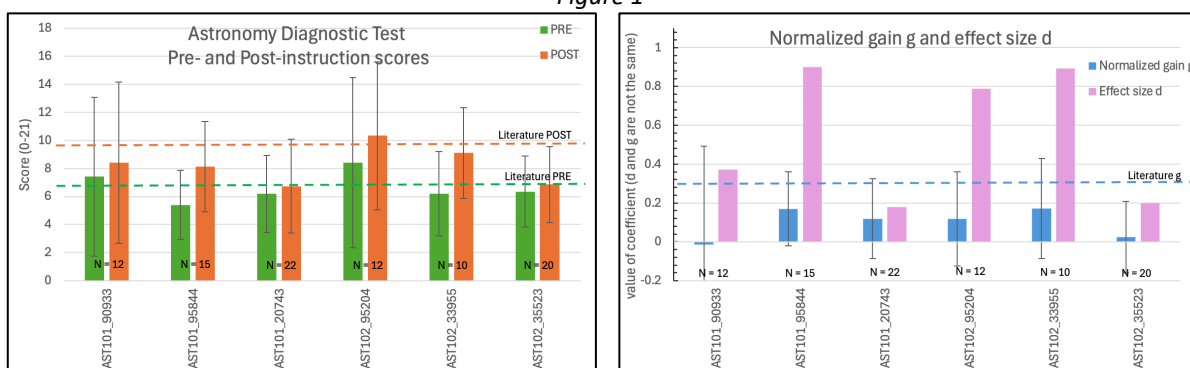
Summary and interpretation of the findings. What evidence is there that the learning outcome is being met?

This is the first time we are using ADT to assess general knowledge in astronomy (PLO#1). Our goal is to obtain scores similar or better than those cited the literature. The table below summarizes our comparative results this year (the uncertainties are standard deviations):

	SCCC result	Literature	Comparisons
Total number of students for lab+lecture format	91 matched	Pre: 1730 Post: 1371	⚠ caution
Mean Pre-score	6.6 ± 3.8	6.7 ± 1.1	✓ OK
Mean Post-score	8.0 ± 4.0	9.8 ± 1.9	✓ OK
Mean Normalized Gain, g	0.09 ± 0.27	0.21 ± 0.10	⚠ caution
Mean Effect Size, d	0.55 ± 0.34	$\sim 1.63 \pm 0.04$	⚠ caution

Figure 1 below depicts the distributions of scores and coefficients d and g for each section of AST101 and AST102. The error bars are the standard deviation of the samples.

Figure 1



Evidence that the learning outcome is being met:

- Our average Pre- and Post-scores (Pre: 6.6 ± 3.8 vs 6.7 ± 1.1 , and Post: 8.0 ± 4.0 vs 9.8 ± 1.9) are within the literature values when taking into consideration our $\pm$ standard deviation.
- Our effect size $d = 0.55$ is in the range of medium values (~ 0.5), it measures the impact in our student's learning, and is an index known to be more reliable for smaller samples like ours. Overall, we interpret $d = 0.55 \pm 0.34$ as a reasonable first result but would like to improve it to a large (~ 0.8) effect size in the future. The high value of d from literature is mostly due of their smaller standard deviation (± 0.04), in contrast to the large standard deviation of our sample (± 0.34). We think our large standard deviation, which is present in all our measurements, is due in part to the non-uniformity of student mathematical backgrounds (more below).

Issues we need to consider when comparing our results to national averages:

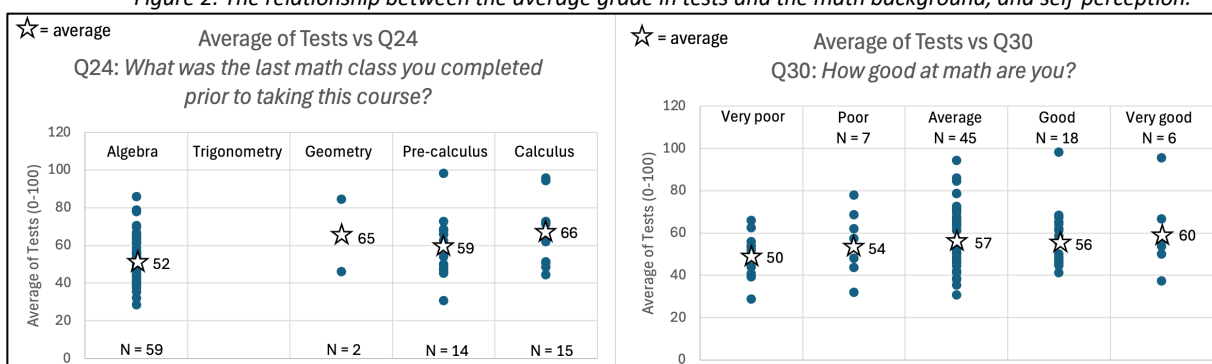
- Low statistics:
The low number of students in our sample compared to the high numbers in the literature prevents a reliable comparison of parameters that strongly depend on statistics, such as the mean normalized gain g. The uncertainty of statistical measurements scales as $1/\sqrt{\text{Number of students}}$. Our sample is about 15 times smaller than the sample in the literature, which means our uncertainties will be about 4 times greater than the literature.

We plan to apply ADT to more classes in future semesters, aggregate the data and perform improved comparisons for the next assessment report.

- Non-uniformity of math backgrounds and low academic motivation:
Most students included in the literature are non-science majors with university-level math prerequisites (i.e., algebra and trigonometry; equivalent to MAT120 at SCCC). At SCCC, the mathematical prerequisite for AST101 and AST102 is basic algebra MAT007. Thus, because ADT places some emphasis in math in a few of its questions (Q3, Q15, and order of magnitude questions Q11, Q20; thus 4 out of 21 questions), this is poised to present a problem for those students with math anxiety and/or limited math skills. We therefore would expect somewhat lower ADT scores due to this fact.

Question 24 in ADT asks, “What was the last math class you completed prior to taking this course?”. And Question 30 asks, “How good at math are you?”. We plotted the answers to these questions against the post-test scores, mean normalized gain, final grade and test averages, with only test averages showing a noticeable pattern (please refer to the Figure 2 below): students who have taken math beyond MAT007 or see themselves as “good at math” perform better in AST101 and AST102 tests by a factor of about 10 points, on average (the average values are represented by stars in the figure). The two questions seem to have the same relationship with the test averages.

Figure 2: The relationship between the average grade in tests and the math background, and self-perception.



Based on these graphs, we think that it is not only the varied mathematical background that increases the standard deviation in our samples, but also the perception of how good students see themselves at math.

Why do the lecture test correlate better with math-level/self-perception? Interestingly, there are less than 10% of math-related questions in the multiple-choice *lecture* tests applied to AST101 and AST102. Most mathematical training and questioning are done during the 13 available labs, which are completed in groups in class, with 3 or more submission chances per question, and further dropping the 3 lowest lab grades at the end of the semester. Most students achieve a letter grade A in labs, which raises the overall final grade of the course to a minimum plateau, making the final grade less sensitive to tracing individual student effort, or anxiety to math. Thus, given the organization of the courses in this report, the test grades are the best tracers of individual student effort: the effort to study the (virtually non-mathematical) concepts described during lecture. We think we can equate this type of effort to academic motivation, assuming that all students have sufficient and equivalent reading and understanding skills. We believe the correlations in Figure 2 are telling us that the most

academically motivated students in AST101 and AST102, who would strive for better grades and do not settle for only a passing grade, are also those who took math beyond MAT007.

Supporting this interpretation of academic motivation is the fact that less than half of the students in each class takes the optional final test. The optional final test is a selection of questions copied verbatim from the previous three multiple-choice lecture tests, for which all the students have the answer keys. There is no penalty for taking the final, and if its grade is the lowest of all tests, it is then dropped before the final grade calculation. Thus, all students could improve their grades tremendously by taking the final if they would study their previous tests. The conditions are explained since the first day of class. Invariably, very few students even attempt the final test, and most who attempt do not make an effort to study as shown by their guessing scores, missing a relatively easy opportunity to improve their course grade. Additionally, several different generous extra credit opportunities are offered during the semester, and only a handful of students take advantage of those as well. For anyone might believe the solution is to tighten the grade rules, this is incorrect as science entry-level classes are often “being shopped” by students, seeking the easiest way to achieve a passing grade. Tightening the rules substantially would empty our courses and fill in alternative courses.

We note that no clear correlation was found when comparing Q24 and Q30 with post-scores, normalized gain or final grade.

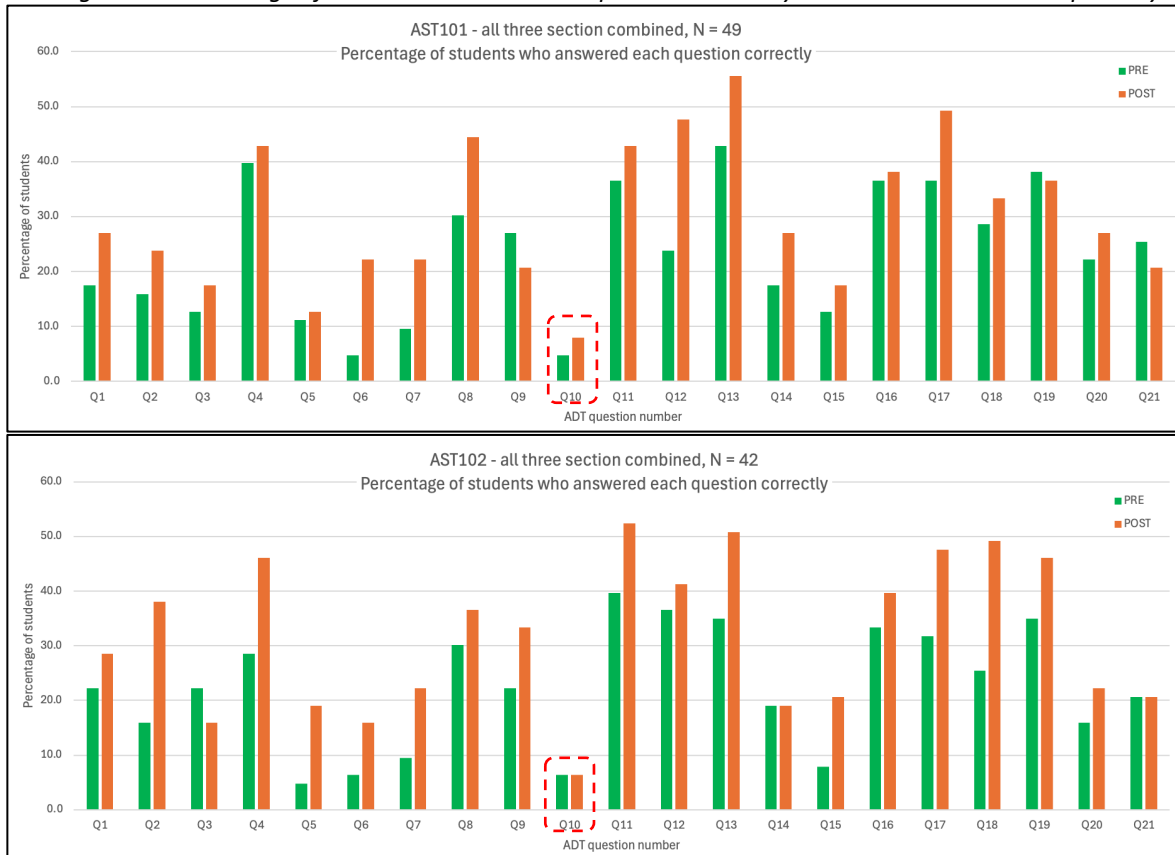
Perhaps the reason for a larger spread and no clear correlation of those questions with the ADT scores is due to the fact that the commitment to answer survey questions whose result will not count in their final grade – as ADT was presented to students, is an additional source of variability in our data. For example, one of the students in the current sample had an ADT pre-score of 19, and got A’s in all her AST101 tests. I remember she seemed to be rushed to finish the last installment of the assessment for some reason, and indeed got a post-score of only 16. This represents a normalized gain of -1.5, an outlier data point, which tremendously skewed our average gain for her class. This brilliant student became a physics and astronomy tutor in a subsequent semester, thus per post-score is truly not a reliable data point.

Concerning the lack of correlation with the normalized gain, the index g is known to be skewed toward low-pre scores from its formula, and not ideal for a small sample like ours. Lastly, the final grades include labs and homework assignments, which are kept as low-stake assignments where all students can achieve top grades, thus hiding some of the purely individual effort/high-academic motivation aspects of a grade.

- We must seek other community college data for comparison:
Yet, even amongst the group of students with a math background higher than MAT007, the test grades vary significantly. This may be endemic to the characteristics of community college students, who often juggle study, work, family and financial responsibilities at a level much greater than the four-year institution students polled in the literature. It is the case that a community college student with sound academic motivation sometimes needs to settle for a lower score to be able to take care of other more demanding aspects of their life. Therefore, we believe it is important to compare our results with other community college results, because then we will be comparing similar student populations. We will be seeking ADT community college data in the future to improve our analysis.

This was the first time we applied ADT to our classes, and it is useful to other instructors applying the same test to know that there were no significant differences in the performance of AST101 vs AST102.

Figure 3 – Percentage of students who answered questions correctly in AST101 and AST102 separately.



We were particularly concerned with the performance of AST102 students in questions 18 and 19 that refer to lunar phases - a topic that is not covered extensively AST102, but is in AST101. When subtracting these questions from AST102 average scores, we found no major difference. In fact, many AST102 students answered Q18 and Q19 correctly in the pre-test, perhaps due to previous knowledge.

All the remaining questions in ADT are equally covered during the first half of the semester for both AST101 and AST102: starting with an overview of astronomy and the main components of our expanding universe, and then focusing on scales, coordinates, motions in the sky, seasons, properties of light, gravity and orbits, and the Sun. The courses diverge during the second half, with AST101 focusing on the solar system (the Moon, planets and natural satellites, comets, asteroids, Kuiper Belt), and AST102 focusing on stellar evolution, galaxies and cosmology. For example, after the initial overview of astronomy at the start of the semester, even AST101 students received enough instruction to address question 16 on the topic of cosmology: “Q16: According to modern ideas and observations, what can be said about the location of the center of the Universe?”.

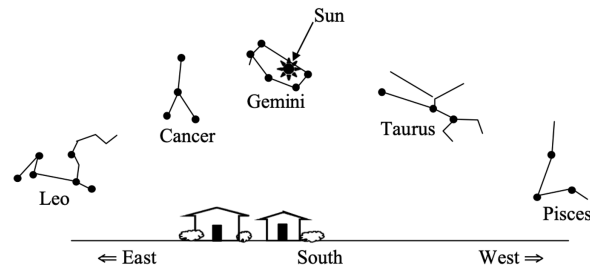
As can be seen in Figure 3, question 10 was the most challenging for all students in this sample:

10. If you could see stars during the day, this is what the sky would look like at noon on a given day. The Sun is near the stars of the constellation Gemini. Near which constellation would you expect the Sun to be located at sunset?

A. Leo
B. Cancer

C. Gemini
D. Taurus

E. Pisces



Although students have explored the motion of the Sun in the sky with the software Stellarium, also during planetarium demonstrations, lectures and a lab on the motion of the Sun along the plane of the ecliptic that is underscored by the zodiacal constellations (including questions such as “In what constellation is the Sun on your birthday? Check with Stellarium”), students are indeed rarely exposed to questions about what would happen next after a given interval of time, which forces them to absorb the ideas in different ways. These types of questions will now be added to their labs more often. However, it is important for instructors applying any research-based assessment instrument not to teach students preparing them to answer the exact questions in the assessment. This would defeat the purpose of all the work done by our predecessors in educational research.

How will assessment results be shared with program faculty, staff and other stakeholders?

The assessment report will be emailed to all astronomy faculty in the three campuses and to our department chair. Assessment results are often presented at the department meetings as well.

Actions required to improve teaching and learning in light of the findings. Who will be responsible for overseeing any actions?

- Increase astronomy faculty awareness about standardized assessment, and the benefits of improved statistics. Invite more faculty to collect ADT data. Responsible: Glenda Denicolo.
- Emphasize to students that all questions in the assessment must receive an answer, even if they are not sure or have no idea what the answer should be, which is often the case during pre-tests. There should be no abstentions. When students leave questions blank the remaining parts of their scores cannot be used in a full pre-/post-test comparison. Responsible: all who apply ADT in their courses.

Description and timeline for follow-up activities.

During the summer break:

- Remote meeting with the entire astronomy faculty to explain the implementation of standardized assessment, and the need to increase our statistics.
- Begin the process of requesting ADT data from community colleges during the summer meeting of the American Association of Physics Teachers in August 2025.

Two weeks before start of a semester:

- Email entire astronomy faculty about participating in standardized assessment, in order to increase our statistics.

Section III: Report on Program Review Recommendations

What is the date of the last program review?	Last Program Review was <u>March 2021</u> .
Please list all of the recommendations that were made and indicate what you have done to address each recommendation and how you are making progress. If you are not making progress, explain why.	This item has been addressed in previous reports. There are no more recommendations in astronomy yet to be met.

Please send any material associated with the assessment (Excel spreadsheet/ rubric/ reports, etc.) with this report to Lauren Tacke-Cushing, Associate Vice President of Academic Affairs @ tackel@sunysuffolk.edu and to your campus Associate Dean of Academic Affairs. Thank you.

How to obtain the Astronomy Diagnostic Test and its answer key?

- 1) Register as a teacher in <https://www.physport.org/>
- 2) Once registration is confirmed, go to <https://www.physport.org/assessments/> and search for assessment in Astronomy. ADT should be one of the first listed. Click on it.
- 3) You will be instructed on how to download the test. In the process, you will read security warnings to prevent incorrect use or distribution of any research-based assessment instrument. Also, there will be instructions on how to present this assessment to students as a survey whose result will not affect their grades, and to emphasize the importance of their honest participation.

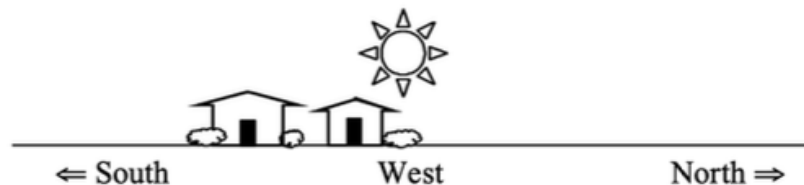
A copy of the pre-test is attached below. Questions 6 and 11 were modified from the original ADT test to refer to the International Space Station instead of the space shuttle (outdated).

Introductory Astronomy Survey

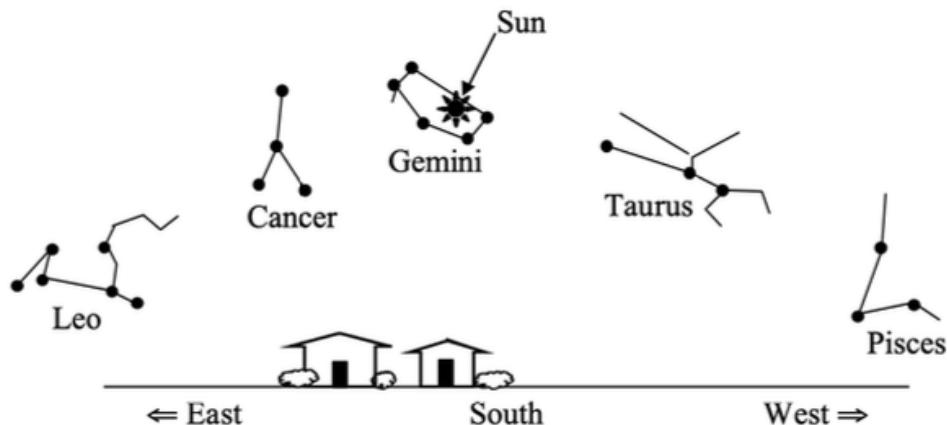
1. As seen from your current location, when will an upright flagpole cast no shadow because the Sun is directly above the flagpole?
 - A. Every day at noon.
 - B. Only on the first day of summer.
 - C. Only on the first day of winter.
 - D. On both the first days of spring and fall.
 - E. Never from your current location.
2. When the Moon appears to completely cover the Sun (an eclipse), the Moon must be at which phase?
 - A. Full
 - B. New
 - C. First quarter
 - D. Last quarter
 - E. At no particular phase
3. Imagine that you are building a scale model of the Earth and the Moon. You are going to use a 12-inch basketball to represent the Earth and a 3-inch tennis ball to represent the Moon. To maintain the proper distance scale, about how far from the surface of the basketball should the tennis ball be placed?
 - A. 4 inches ($\frac{1}{3}$ foot)
 - B. 6 inches ($\frac{1}{2}$ foot)
 - C. 36 inches (3 feet)
 - D. 30 feet
 - E. 300 feet
4. You have two balls of equal size and smoothness, and you can ignore air resistance. One is heavy, the other much lighter. You hold one in each hand at the same height above the ground. You release them at the same time. What will happen?
 - A. The heavier one will hit the ground first.
 - B. They will hit the ground at the same time.
 - C. The lighter one will hit the ground first.
5. How does the speed of radio waves compare to the speed of visible light?
 - A. Radio waves are much slower.
 - B. They both travel at the same speed.
 - C. Radio waves are much faster.
6. Astronauts inside the International Space Station (ISS) float around as it orbits the Earth because
 - A. there is no gravity in space.
 - B. they are falling in the same way as the ISS.
 - C. they are above the Earth's atmosphere.
 - D. there is less gravity inside the ISS.
 - E. more than one of the above.

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7. Imagine that the Earth's orbit were changed to be a perfect circle about the Sun so that the distance to the Sun never changed. How would this affect the seasons?
- We would no longer experience a difference between the seasons.
 - We would still experience seasons, but the difference would be much LESS noticeable.
 - We would still experience seasons, but the difference would be much MORE noticeable.
 - We would continue to experience seasons in the same way we do now.
8. Where does the Sun's energy come from?
- The combining of light elements into heavier elements
 - The breaking apart of heavy elements into lighter ones
 - The glow from molten rocks
 - Heat left over from the Big Bang
9. On about September 22, the Sun sets directly to the west as shown on the diagram below. Where would the Sun appear to set two weeks later?
- Farther south
 - In the same place
 - Farther north



10. If you could see stars during the day, this is what the sky would look like at noon on a given day. The Sun is near the stars of the constellation Gemini. Near which constellation would you expect the Sun to be located at sunset?
- Leo
 - Cancer
 - Gemini
 - Taurus
 - Pisces



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11. Compared to the distance to the Moon, how far away is the International Space Station from Earth?
 - A. Very close to the Earth
 - B. About half way to the Moon
 - C. Very close to the Moon
 - D. About twice as far as the Moon

12. As viewed from our location, the stars of the Big Dipper can be connected with imaginary lines to form the shape of a pot with a curved handle. To where would you have to travel to first observe a considerable change in the shape formed by these stars?
 - A. Across the country
 - B. A distant star
 - C. Europe
 - D. Moon
 - E. Pluto

13. Which of the following lists is correctly arranged in order of closest-to-most-distant from the Earth?
 - A. Stars, Moon, Sun, Pluto
 - B. Sun, Moon, Pluto, stars
 - C. Moon, Sun, Pluto, stars
 - D. Moon, Sun, stars, Pluto
 - E. Moon, Pluto, Sun, stars

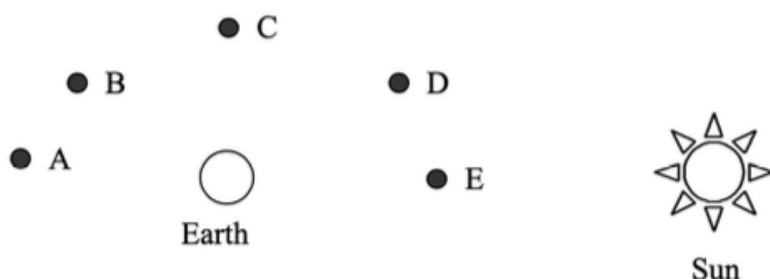
14. Which of the following would make you weigh half as much as you do right now?
 - A. Take away half of the Earth's atmosphere.
 - B. Double the distance between the Sun and the Earth.
 - C. Make the Earth spin half as fast.
 - D. Take away half of the Earth's mass.
 - E. More than one of the above

15. A person is reading a newspaper while standing 5 feet away from a table that has on it an unshaded 100-watt light bulb. Imagine that the table were moved to a distance of 10 feet. How many light bulbs in total would have to be placed on the table to light up the newspaper to the same amount of brightness as before?
 - A. One bulb.
 - B. Two bulbs.
 - C. Three bulbs.
 - D. Four bulbs.
 - E. More than four bulbs.

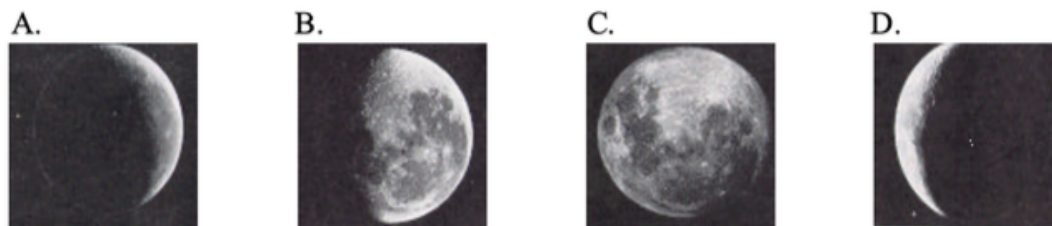
16. According to modern ideas and observations, what can be said about the location of the center of the Universe?
 - A. The Earth is at the center.
 - B. The Sun is at the center.
 - C. The Milky Way Galaxy is at the center.
 - D. An unknown, distant galaxy is at the center.
 - E. The Universe does not have a center.

17. The hottest stars are what color?
 - A. Blue
 - B. Orange
 - C. Red
 - D. White
 - E. Yellow

18. The diagram below shows the Earth and Sun as well as five different possible positions for the Moon. Which position of the Moon would cause it to appear like the picture at right when viewed from Earth?



19. You observe a full Moon rising in the east. How will it appear in six hours?



Images © UC Regents/Lick Observatory. Unauthorized use prohibited.

20. With your arm held straight, your thumb is just wide enough to cover up the Sun. If you were on Saturn, which is 10 times farther from the Sun than the Earth is, what object could you use to just cover up the Sun?
- Your wrist
 - Your thumb
 - A pencil
 - A strand of spaghetti
 - A hair
21. Global warming is thought to be caused by the
- destruction of the ozone layer.
 - trapping of heat by nitrogen.
 - addition of carbon dioxide.
22. In general, how confident are you that your answers to this survey are correct?
- Not at all confident (just guessing)
 - Not very confident
 - Not sure
 - Confident
 - Very confident

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23. What is your college major (or current area of interest if undecided)?
- A. Business
 - B. Education
 - C. Humanities, Social Sciences, or the Arts
 - D. Science, Engineering, or Architecture
 - E. Other
24. What was the last math class you completed prior to taking this course?
- A. Algebra
 - B. Statistics
 - C. Trigonometry
 - D. Pre-Calculus
 - E. Calculus
25. What is your age?
- A. 0-20 years old
 - B. 21-23 years old
 - C. 24-30 years old
 - D. 31 or older
 - E. Decline to answer
26. Which best describes your home community (where you attended high school)?
- A. Rural
 - B. Small town
 - C. Suburban
 - D. Urban
 - E. Not in the USA
27. What is your gender?
- A. Female
 - B. Male
 - C. Decline to answer
28. Which best describes your ethnic background?
- A. African-American
 - B. Asian-American
 - C. Native-American
 - D. Hispanic-American
 - E. None of the above (see question 29 below)
29. Which best describes your ethnic background?
- A. African (not American)
 - B. Asian (not American)
 - C. White, non-Hispanic
 - D. Multicultural
 - E. None of the above (see question 28 above)

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30. How good at math are you?
- A. Very poor
 - B. Poor
 - C. Average
 - D. Good
 - E. Very good
31. How good at science are you?
- A. Very poor
 - B. Poor
 - C. Average
 - D. Good
 - E. Very good
32. Which best describes the level of difficulty you expect/experienced from this course?
- A. Extremely difficult for me
 - B. Difficult for me
 - C. Unsure
 - D. Easy for me
 - E. Very easy for me
33. How many astronomy courses at the college level have you taken?
- A. I'm re-taking this course.
 - B. This is my first college-level astronomy course.
 - C. This is my second college-level astronomy course.
 - D. I've completed more than two other college-level astronomy courses.

END OF SURVEY